INCH-POUND

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PERFORMANCE SPECIFICATION
FOR
REFRIGERATOR, MECHANICAL, FIELD, PORTABLE WALK-IN, PLUG-IN
(150 CUBIC FEET)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a portable 150-cubic foot capacity walk-in, plug-in, field refrigerator used with a 5000 British thermal unit (Btu) capacity refrigeration unit (see 6.1)

1.2 Part identification number (PIN). The PIN will consist of the basic number of the specification and the appropriate PIN structure as specified in 6.7.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development and Engineering Center, Natick, MA 01760-5018 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4110

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents specified in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

A-A-52128 - Shelving, Storage, Stationary and Mobile, Food Storage

DEPARTMENT OF DEFENSE

MIL-C-46168 - Coating, Aliphatic Polyurethane, Chemical Agent Resistant

MIL-C-53039 - Coating, Aliphatic Polyurethane, Single Component, Chemical Agent Resistant

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STANDARDS

FEDERAL

FED-STD-595 - Colors Used in Government Procurement

DEPARTMENT OF DEFENSE

MIL-STD-810 - Environmental Test Methods and Engineering Guidelines

(Unless otherwise indicated, copies of the above specifications and standards are available from the Defense Automated Printing Service, Building 4D, Robbins Avenue, Philadelphia, PA 19111-5094).

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

UNDERWRITERS LABORATORIES INC. (UL)

Standard 98 - Enclosed and Dead-Front Switches

(Application for copies should be addressed to the Underwriters Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062-2096)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

Standard No. 70 - National Electrical Code

(Application for copies should be addressed to the National Fire Protection Association, One Battery March Park, Quincy, MA 02269-9101).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM F883 - Standard Performance Specification for Padlocks

(Application for copies should be addressed to the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohoken, PA 19428-2959).

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NSF INTERNATIONAL

Standard No. 7 - Food Service Refrigerators and Storage Freezers

(Application for copies should be addressed to NSF International, 3475 Plymouth Road, P. O. Box 1468, Ann Arbor, MI 48106).

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.1.1.

3.2 **Operating requirements**. Refrigerators shall satisfy the following minimum functional, operational and performance requirements.

3.2.1 Heat leakage. Heat infiltration or loss through the refrigerator shall be minimized.

3.2.2 Safety. The refrigerator shall minimize flammability, operational, toxic and personnel hazards.

3.2.3 Drainage and watertightness. The refrigerator shall be watertight with the door closed and capable of being drained with the door open.

3.2.4 Door. The door shall be located on the front of the refrigerator, open outward, and be hinged on the right hand side when facing the door. The door opening shall be a minimum 61 inches high and 37 inches wide. Door hardware shall not damage the exterior of the refrigerator when the door is opened. External hardware shall be protected from environmental and physical damage. Means shall be provided to hold the door in the full open position.

3.2.5 Door handle and latch. The door shall have an external handle which shall operate a locking device to hold the door in

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the closed position during all modes of shipment and rough handling. The latch shall have provisions to accept the padlock specified in 3.5.2. With the padlock in place, the door shall be incapable of being opened from the outside. The door shall have an internal release handle to allow the door to open from the inside to permit a person in the refrigerator to exit when the refrigerator is padlocked on the outside. The handle and latch shall not be damaged during shipment.

3.2.6 Electric power. Unless otherwise specified (see 6.2), all electrical devices and components shall be capable of operating on a 120 volt, 60 hertz, single phase power.

3.2.6.1 Interior light. The refrigerator shall have a vapor proof lighting fixture mounted on the interior and a recessed or protected vapor proof switch located near the door on the exterior of the refrigerator. The fixture shall have a protective cover which is shatterproof and removable without tools for bulb replacement. The fixture shall be furnished with a 100 watt, 120 volt, incandescent bulb appropriate for use in a refrigerator.

3.2.6.2 Pilot light. The refrigerator shall have a pilot light enclosed in a vapor-tight housing which illuminates red when the interior light switch is in the "on" position. The pilot light shall be located externally adjacent to the door and shall be protected from damage.

3.2.6.3 Plug and receptacle. A weather-tight 15 amp receptacle, designed for 3-wire, 2-pole grounded service shall be provided. The receptacle shall be of the recessed male, locking type. The receptacle shall be located externally, adjacent to the door. A cap or flap cover shall be provided to seal the receptacle housing when not in use. The plug shall form a weathertight seal when inserted into the receptacle and both the plug and cap or flap cover shall be captive to the refrigerator to prevent loss.

3.2.7 Temperature indicator. A temperature indicator, with a range of -25^{0}F to $+125^{0}$F, shall be provided and installed on the exterior wall near the door. The sensing element shall provide a mean temperature reading of the refrigerator. The temperature indicator and sensing element shall be protected from physical damage.

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3.2.8 Ozone-depleting and hazardous substances and toxic chemicals. Class I ozone-depleting substances, hazardous substances and toxic chemicals, as determined by the Environmental Protection Agency, shall not be contained in, and shall be prohibited from use during the fabrication of the refrigerators (see 6.6).

3.3 **Support or ownership requirements**. The refrigerator shall meet the following support or ownership requirements.

3.3.1 Storage capacity. The refrigerator shall have a minimum internal volume of 150 cubic feet with the refrigeration unit in place.

3.3.2 Transportability. The refrigerator shall have an integral forklift provisions and shall also have corner fittings or lifting eyes located on the top of the unit.

3.3.3 Floor racks. Removable floor racks, sized to completely cover the internal floor area of the refrigerator, shall be provided.

3.3.4. Evaporator opening cover. A removable and reusable weathertight cover shall be provided to seal the opening in the rear endwall when the refrigeration unit is not in place.

3.3.5 Shelving. When specified (see 6.2)shelving conforming to A-A-52128 shall be furnished in the required type, style, size and quantity.

3.3.6 Tie-down assemblies. When specified (see 6.2), a minimum of 4 internal tie-down assemblies (2 upper and 2 lower) shall be furnished on each sidewall.

3.3.7 Marking. Identification plates shall be corrosion resistant and permanently attached to the outside of the refrigerator door. Numbering and lettering shall be permanently inscribed and shall be no less than 1/16 inch high. All plates shall blend with the refrigerator finish and shall be nonreflective.

3.3.7.1 Identification plate data. The identification plate shall contain the following information:

1. U. S.
2. Military model number and nomenclature.
3. National stock number (NSN)

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4. Serial number.
5. Contract number.
6. Manufacturer's name and address.
7. Manufacturer's model number.
8. Commercial and Government Entity (CAGE) number.
9. Technical manual number (see 6.2 and 6.5).
10. Weight.

3.3.7.2 Marine Corps identification plate. When the procurement is for the Marine Corps (see 6.2), the identification plates, together with any data that differ from the foregoing (see 3.3.7.1), shall be obtained from the Commandant of the Marine Corps (Code CS6), Headquarters, Marine Corps, Washington, DC 20380.

3.3.8 Finish. Except where detrimental to proper operation, external surfaces of the refrigerator shall have a nontoxic chemical agent resistant finish. The exterior surfaces of the refrigerator shall be painted color camouflage green or lusterless dark green. The word "CARC", together with the month and year of finishing, shall be stencilled near the identification plate in block letter not less than one-inch high using lusterless black chemical agent resistant paint.

3.3.9 Interchangeability. All replaceable component parts of the refrigerators shall be interchangeable with the same parts of other refrigerators furnished under the same contract.

3.3.10 Sanitary requirements. The refrigerator shall be designed to facilitate sanitizing.

3.4 **Environmental requirements**. The refrigerator shall meet the following environmental requirements.

3.4.1 Shock and vibration. The refrigerator shall be structurally sound.

3.4.2 Weatherproof. The refrigerator shall be weatherproof.

3.4.3 Storage. The refrigerator shall withstand temperature extremes.

3.4.4 Corrosion resistance. The refrigerator shall withstand exposure to a salt fog atmosphere.

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3.4.5 Grounding for Navy. For Navy procurements (see 6.2), provisions shall be made to permanently ground and bond all exposed internal and external metal surfaces of the refrigerator floor, ceiling, door and walls to a grounding rod.

3.5 **Interface and interoperability requirements**. The refrigerator shall meet the following interface and interoperability requirements.

3.5.1 Refrigerator interface. The refrigerator shall be designed for use with 5000 Btu diesel engine and electric motor driven refrigeration units (see 6.1). An opening, 28-7/8 inches square shall be made in the endwall opposite the door to accept the refrigeration unit. The opening shall be centered along the width of the panel and the bottom of the opening shall be no less than 14 inches nor more than 15 inches above the bottom edge of the panel. Provisions shall also be incorporated on the panel to allow fastening of the refrigeration unit to the refrigerator.

3.5.2 Padlock. The refrigerator shall be supplied with a captive padlock conforming to ASTM F883 and two keys.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.1.1)
b. Conformance inspection (see 4.1.2)

4.1.1 First article inspection. When a first article is required (see 3.1 and 6.2), a sample shall be examined for all the verifications listed in table I.

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TABLE I. Verification methods

Title	Require-ment	Verification	Title	Require-ment	Verification
OPERATING REQUIREMENTS	3.2	4.3.2	Shelving	3.3.5	4.3.3.5
Heat leakage	3.2.1	4.3.2.1	Tie-down assemblies	3.3.6	4.3.3.6
Safety	3.2.2	4.3.2.2	Marking	3.3.7	4.3.3.7
Drainage and watertightness	3.2.3	4.3.2.3	Identification plate data	3.3.7.1	4.3.3.7.1
Door	3.2.4	4.3.2.4	Marine Corps identification plate	3.3.7.2	4.3.3.7.2
Door handle and latch	3.2.5	4.3.2.5	Finish	3.3.8	4.3.3.8
Electric power	3.2.6	4.3.2.6	Interchangeability	3.3.9	4.3.3.9
Interior light	3.2.6.1	4.3.2.6.1	Sanitary requirements	3.3.10	4.3.3.10
Pilot light	3.2.6.2	4.3.2.6.2	**ENVIRONMENTAL REQUIREMENTS**	3.4	4.3.4
Plug and receptacle	3.2.6.3	4.3.2.6.3	Shock and vibration	3.4.1	4.3.4.1
Temperature indicator	3.2.7	4.3.2.7	Weatherproof	3.4.2	4.3.4.2
Ozone-depleting and hazardous substances and toxic chemicals	3.2.8	4.3.2.8	Storage	3.4.3	4.3.4.3
SUPPORT OR OWNERSHIP REQUIREMENTS	3.3	4.3.3	Corrosion resistance	3.4.4	4.3.4.4
Storage capacity	3.3.1	4.3.3.1	Grounding for Navy	3.4.5	4.3.4.5
Transportability	3.3.2	4.3.3.2	**INTERFACE AND INTEROPERABILITY REQUIREMENTS**	3.5	4.3.5
Floor racks	3.3.3	4.3.3.3	Refrigerator interface	3.5.1	4.3.5.1
Evaporator opening cover	3.3.4	4.3.3.4	Padlock	3.5.2	4.3.5.2

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4.1.2 Conformance inspection. Conformance inspection of the refrigerators shall consist of all the verifications listed in table I. Sampling for this inspection shall be as specified in the contract or order (see 6.2)

4.2 Order of verifications. Verifications listed in table I can be performed in any order except verification tests shall be conducted in the order shown in table II. The first weatherproof test shall be conducted before the application of the finish to the refrigerator. All other verifications listed in table I shall be done after finishing is completed. A second weatherproof test shall be done upon completion of all other testing as indicated in table II.

TABLE II. Verification test sequence

Test Sequence	Requirement Paragraph	Characteristic	Verification Paragraph
1	3.4.2	Weatherproof*	4.3.4.2
2	3.4.1	Shock and vibration	4.3.4.1
3	3.3.2	Transportability	4.3.3.2
4	3.2.3	Drainage and watertightness	4.3.2.3
5	3.4.3	Storage	4.3.4.3
6	3.4.4	Corrosion resistance	4.3.4.4
7	3.2.1	Heat leakage	4.3.2.1
8	3.4.2	Weatherproof*	4.3.4.2

* See paragraph 4.2

4.3. Verification methods. Verification methods can include visual examination, measurement, testing, simulation, modeling, engineering evaluation, component properties analysis, certification and similarity to previously-approved or previously-qualified designs.

4.3.1 Verification alternatives. The contractor may propose alternative test methods, techniques, or equipment, including the application of statistical process control, tool control, or cost-effective sampling procedures to verify performance (see 6.4).

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4.3.2 **Operating requirements verification**. All operating requirements shall be verified as follows.

4.3.2.1 Heat leakage test. A refrigeration unit or simulation thereof shall be installed in the refrigerator for this test. There shall be a minimum five-foot clearance from the test room walls. The overall heat leakage shall be determined as follows:

a. An electrical fan, of sufficient capacity to ensure that no uncirculated air pocket will exist, shall be installed inside the refrigerator. The electrical energy of the fan motor shall be measured by a watt-hour meter with a scale of 0.1 kilowatt-hour or less and this heat equivalent shall be added to other sources of heat added to the refrigerator.

b. Four thermocouples, shielded from the effects of radiant heat, shall be installed inside the refrigerator and positioned within 6 inches of the ceiling and floor at two opposite corners. The interior temperature shall be taken as the average of the four thermocouples.

c. Heating elements shall be installed approximately in the center of the refrigerator. Electrical heat input shall be measured by a watt-hour meter with a scale increment of 0.1 kilowatt-hour or less.

d. Eight thermometers or thermocouples, shielded from the effects of radiant heat, shall be installed approximately 18 inches from each corner of the refrigerator in the room where it is placed. The ambient temperature shall be taken as an average of these eight thermocouple readings.

e. All openings in the refrigerator shall be sealed and the refrigerator shall remain undisturbed throughout the test.

f. The ambient temperature shall then be maintained constant within $\pm$ 2^0F. Sufficient, steady, non-varying heat shall be added to the interior of the refrigerator to maintain a minimum of 60°F and a maximum of 100°F temperature difference between the interior and exterior of the refrigerator.

g. The test shall be considered concluded when eight consecutive half-hourly readings indicate that:

(1) The ambient temperature has been maintained constant within $\pm$ 2°F.

(2) The temperature inside the refrigerator has been maintained constant within $\pm$ 1°F.

h. The overall heat leakage coefficient U of the refrigerator is given by:

$$U = \frac{H_1}{(T_1 - T_2)}$$

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H_1= Sum of all electrical heat energies added to the refrigerator (Btu/hr)
T_1= Average interior temperature of the refrigerator (100°F to 155°F)
T_2= Average ambient temperature (60°F to 100°F)

i. The heat leakage coefficient shall be 15 Btu/hr/°F or less.

4.3.2.2 Safety certification and inspection. Safety requirements shall be verified as follows.

4.3.2.2.1 Wiring certification. Wiring practices used in the refrigerator shall be certified as conforming to NFPA Standard No. 70 by submission of the manufacturer's certification, a certified test report or appropriate labels and listing marks.

4.3.2.2.2 Electrical components certification. Electrical components shall be certified as conforming to the requirements of UL Standard 98 by submission of the manufacturer's certification, a certified test report or appropriate labels and listing marks.

4.3.2.2.3 Shelving certification When applicable, the contractor shall certify that the shelving conforms to the sanitation requirements of A-A-52128.

4.3.2.3 Drainage and watertightness test Verify watertightness with the door shut by completely covering the floor with water to a depth of not less than 3 inches. Duration of the test shall not be less than 8 hours. Upon completion of the test, there shall be no evidence of water leakage. Verify complete drainage can be accomplished to the outside through the open refrigerator door with the refrigerator located on a level surface.

4.3.2.4 Door inspection. Visually inspect the door to verify it is located on the front of the refrigerator. Verify the door opens outward, is hinged on the right hand side when facing the door and has a means to be held in the full open position. Verify the presence of a means to prevent door hardware from damaging the exterior of the refrigerator when the door is opened. Measure the door opening to verify the opening is a minimum 61 inches high by 37 inches wide.

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4.3.2.5 Door handle and latch inspection Visually inspect for the presence of an external door handle and latch which accepts a padlock. Verify that the door cannot be opened externally with the latch closed and the padlock in place. Verify that the door can be opened internally when both locked and unlocked on the outside. Verify strength of door locking device by verifying that the door does not open by itself during any rough handling or during testing specified herein.

4.3.2.6 Electric power inspection. Verify that all electrical components and devices operate on the power system specified in the contract or order.

4.3.2.6.1 Interior light inspection. Verify the presence of a vapor proof lighting fixture with a vapor proof switch, having a protective cover which can be removed without tools. Verify the switch is located near the door on the exterior wall and that the switch operates the 100-watt, 120-volt incandescent light bulb, which is appropriate for use in a refrigerator.

4.3.2.6.2 Pilot light inspection. Verify the presence of a vapor proof pilot light and housing on the exterior wall near the door. Verify that the light illuminates red when the interior light switch is in the "on" position with 120-volt power source connected to the container. Verify the pilot light is protected from physical damage.

4.3.2.6.3 Plug and receptacle inspection. Visually inspect the refrigerator for a plug and receptacle located on the exterior near the door. Verify the receptacle is designed for 3-wire, 2-pole grounded service and is rated at 15 amperes. Verify the receptacle is of the recessed male, locking type. Verify that a captive plug or flap cover is provided to seal the receptacle when not in use.

4.3.2.7 Temperature indicator inspection. Verify the presence of a temperature indicator located on the exterior wall near the door. Visually inspect to verify a temperature range of -25°F to +125°F and verify the temperature indicator and sensing element are protected from physical damage.

4.3.2.8 Ozone-depleting and hazardous substances and toxic chemical certification. The contractor shall certify that no ozone-depleting or hazardous substances or toxic chemicals listed in publication SD-14 (see 6.6) are contained in the refrigerator or were used during fabrication of the refrigerator.

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4.3.3 **Support or ownership requirements verification.** All support or ownership requirements shall be verified as follows.

4.3.3.1 Storage capacity measurement. Measure the internal volume of the refrigerator, with a refrigeration unit or simulation thereof in place, to verify a minimum internal volume of 150 cubic feet.

4.3.3.2 Transportability test. The refrigerator, with its floor racks in place and its interior containing an evenly distributed load of 4,000 pounds, shall be lifted, transported and placed in position by a forklift truck without securing the refrigerator in any way to the truck. During the transport, the truck shall be driven over a 4-inch by 4-inch timber so that either the two left or two right tires go over the timber. After the forklift testing, the refrigerator shall be raised by its four lifting eyes or corner fittings and held suspended for 5 minutes. When tie-down assemblies are required by the contract or order, they shall be used to secure the load. Upon completion of this testing, the complete refrigerator and floor racks shall be examined to verify that there has been no damage or distortion.

4.3.3.3 Floor rack inspection. Verify that floor racks are provided and that they can be lifted and positioned by one person, and that the quantity furnished completely covers the internal floor areas of the refrigerator.

4.3.3.4 Evaporator opening cover inspection. Verify the cover is removable, weathertight, reusable and completely seals the evaporator opening in the rear endwall of the refrigerator.

4.3.3.5 Shelving inspection. When shelving is furnished, verify that it is the type, style and size and in the quantity specified in the contract or order.

4.3.3.6 Tie-down assemblies inspection. When tie-down assemblies are required, verify each refrigerator has a minimum of 4 assemblies with 2 located in the upper portion and 2 located in the lower portion of each internal sidewall.

4.3.3.7 Marking inspection. Visually verify that identification plates are permanently attached to the refrigerator door end of the refrigerator; that lettering is legible, permanently inscribed in characters not less than 1/16 inch high and that the plate surface is nonreflective.

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4.3.3.7.1 Identification plate inspection. Visually identify that all information is complete and accurate.

4.3.3.7.2 Marine Corps identification plate inspection. Visually verify that the information is complete and corresponds to that obtained from the Marine Corps, when applicable.

4.3.3.8 Finish certification and examination. The contractor shall certify that the refrigerator's exterior is painted camouflage green 383 or matches color chip number 34082 or 34094 of FED-STD-595, and that the lettering is painted black matching color chip number 37030 of FED-STD-595. Both shall meet the test requirements of MIL-C-53039 or MIL-C-46168(see 6.2). Painted surfaces shall be visually examined to verify that there are no streaks, crops, pinholes, checks, cracks, or missed areas.

4.3.3.9 Interchangeability certification. The contractor shall certify that all component parts of the refrigerator which are replaceable, are interchangeable with the same parts of other refrigerators furnished under the same contract.

4.3.3.10 Sanitary certification The refrigerator shall be certified as conforming to the requirements of NSF Standard No. 7 by submission of the manufacturer's certification, a certified test report or appropriate labels and listing marks.

4.3.4 **Environmental requirements verification**. All environmental requirements shall be verified as follows.

4.3.4.1 Shock and vibration test. The refrigerator with its floor racks in place and containing an evenly distributed load of 4,000 pounds, shall be raised 1 inch above the floor and dropped onto a concrete surface 50 times. Upon completion, visually verify that there has been no damage or distortion to the refrigerator or floor racks.

4.3.4.2 Weatherproof test. The refrigerator, with a refrigeration unit or simulation thereof, shall be sprayed with fresh water at no less than 3 inches per hour at an average downward angle of 45 degrees to simulate rainfall on exposed surfaces of the refrigerator. All sides and the top of the

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refrigerator shall be individually sprayed for 30 minutes with the spray applied perpendicular to the surfaces. Upon completion, a visual verification shall be done to ensure that no water intrusion occurred into the refrigerator or electrical components.

4.3.4.3 Storage test. The refrigerator shall be subjected to a minimum temperature of 155°F for 4 hours, then the room temperature shall be brought to 70°F ± 5°F. After 1 hour, the refrigerator shall then be subjected to a maximum temperature of minus 65°F for 4 hours, then the room temperature increased to 70°F ± 5°F. After 1 hour, inspect the refrigerator and verify that there has been no permanent set or loss of resiliency of packing or gaskets, congealing of lubricants, discoloration, cracking, bulging, checking or crazing.

4.3.4.4 Corrosion resistance test. The refrigerator shall be tested for resistance to the effects of a salt fog atmosphere having a salt concentration of 5 percent ± 1 percent. The refrigerator shall be tested in accordance with MIL-STD-810, Method 509.3, with the refrigerator subjected to 24 hours of salt fog exposure followed by 24 hours of drying time in an ambient temperature of 70°F ± 5°F, followed by 3 more cycles of 24-hour exposure and 24-hour drying time. The refrigerator shall be tested with the door in the open position. Upon completion of testing, visually inspect to verify that there are no signs of corrosion.

4.3.4.5 Grounding for Navy inspection. Visually verify that provisions have been made to permanently ground and bond all exposed external and internal metal surfaces of the refrigerator's floor, ceiling, walls and door to a grounding rod.

4.3.5 **Interface and interoperability requirements verification**. All interface and interoperability requirements shall be verified as follows.

4.3.5.1 Refrigerator interface measurement. Measure the opening in the rear endwall to verify that it is 28-7/8 inches square, centered along the width of the panel and that the bottom of the opening is not less than 14 inches nor more than 15 inches above the bottom edge of the panel. Verify that the refrigerator has provisions to fasten the refrigeration unit to it.

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4.3.5.2 Padlock certification and inspection. The contractor shall certify that the padlock and keys conform to the requirements of ASTM F883. Visually verify that the padlock fits the door latch and is captive to the refrigerator and that both keys unlock the padlock.

5. PACKAGING.

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES
(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.

6.1 Intended use. The refrigerator covered by this specification is intended for worldwide field use by the Military to store and preserve temperature sensitive food items. The refrigerator is military unique since it must operate satisfactorily under climatic conditions encountered throughout the world (e.g. desert, tropic, arctic environments) and in peacetime and wartime operational scenarios (e.g. chemically contaminated environment). The refrigerator is used with the refrigeration units covered by MIL-PRF-12574.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.
b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of documents referenced (see 2.2.1 and 2.3).
c. When a first article is required (see 3.1, 4.1.1 and (6.3).
d. When electric power operation is other than specified (see 3.2.6).

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e. When shelving is required and, if required, type, style, size and quantity (see 3.3.5).
f. When tie-down assemblies are required (see 3.3.6).
g. Technical manual number, when required (see 3.3.7.1).
h. When procurement is for Marine Corps (see 3.3.7.2).
i. When procurement is for Navy (see 3.4.5).
j. Sampling for conformance inspection (see 4.1.2).
k. For Army procurements the CARC finish will meet the requirements of MIL-C-53072 (see 4.3.3.8).
l. Packaging requirements (see 5.1).

6.3 First article. When requiring a first article inspection, contracting documents should provide specific guidance to offerors. This guidance should cover whether the first article is a first article sample, a first production item, or the number of test items. These documents should also include specific instructions regarding arrangements for examinations, approval of first article test results and disposition of first articles. Pre-solicitation documents should provide Government waiver rights for samples for first article inspection to bidders offering a previously-acquired or tested product. Bidders offering such products who wish to rely on such production testing must furnish evidence with the bid that prior Government approval is appropriate for the pending contract.

6.4 Verification alternatives. Contracting documents should provide guidance to offerors regarding the submission of alternatives to specified verification methods (see 4.3.1)

6.5 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, specifications and standards that have been cleared and listed in DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be cited on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract.

6.6 Ozone-depleting and hazardous substances and toxic chemicals. A list of ozone-depleting and hazardous substances and toxic chemicals prohibited from use can be found in publication SD-14 Listing of Toxic Chemicals, Hazardous Substances, and Ozone-Depleting Chemicals. It is available from the Defense Automated Printing Service, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.

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6.7 Part identification number (PIN) structure. The PIN's to be used for items required by this specification are created as follows:

M 12571 - X - X - X - X - X (Example of a PIN: M12571-A1A1A)

- Marking:
  - A - Regular (3.3.7.1)
  - B - Marine Corps (3.3.7.2)
- Grounding:
  - 1 - Regular (3.2.6)
  - 2 - Naval (3.4.5)
- Electrical devices:
  - A - As specified (3.2.6)
  - B - Other than specified
- Tiedown assemblies:
  - 1 - With tiedown assemblies (3.3.6)
  - 2 - Without tiedown assemblies
- Shelving:
  - A - With shelving (3.3.5)
  - B - Without shelving
- Specification number
- M prefix - An item defined by inch-pound units

6.8 Recycled, recovered or environmentally preferable materials. Recycled, recovered or environmentally preferable materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements and promotes economically advantageous life cycle costs.

6.9 Subject term (keyword) listing.

Diesel-powered refrigerator
Electric-powered refrigerator
Food preservation, field
Food storage, field
Refrigeration, field

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6.10 Supersession data. MIL-R-0012571P(GL)and MIL-R-12571N provide a detailed configuration of the refrigerator and are superseded by this performance specification. MIL-R-0012571P(GL) and MIL-R-12571N and associated drawings are available to the contractor on an information basis.

6.11 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:

Army - GL
Navy - YD1
Air Force - 99

Preparing activity:

Army - GL

(Project No. 4110-0502)

Review activities:

Army - CR4, MD1
Navy - MC
Air Force - 35, 82
DLA - IS

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-12571R	**2. DOCUMENT DATE (YYMMDD)** 98-07-10

DOCUMENT TITLE REFRIGERATOR, MECHANICAL, FIELD, PORTABLE, WALK-IN, PLUG-IN (150 CUBIC FOOT)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON *(if applicable)*	**7.DATE SUBMITTED** (YYMMDD)

8. PREPARING ACTIVITY

a. NAME US ARMY NATICK RD&e	b. TELEPHONE *Include Area Code)* (1) Commercial ((508)233-5175 (2) AUTOVON 256-5175
c. ADDRESS *(Include Zip Code)Commander* Commander, US Army natick RD&E Center Attn: SSCNC-WEF Natick, MA 01760-5018	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** DEFENSE QUALITY AND STANDARDIZATION OFFICE 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22401-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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